FRONT WASHER PUMP MOTOR (Continued)

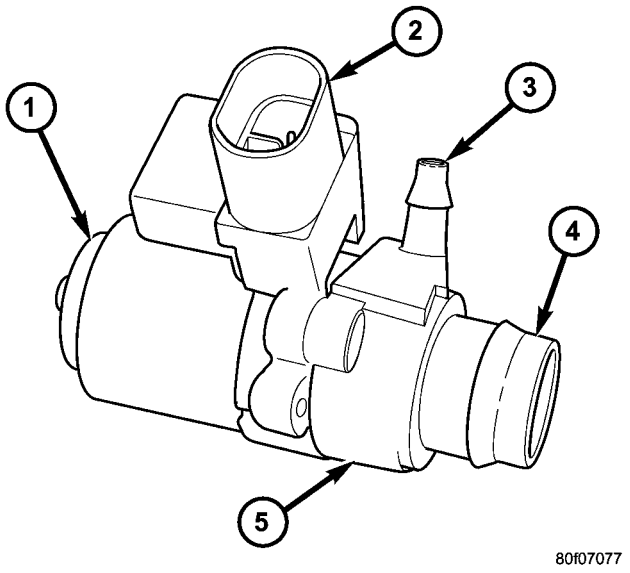

Fig. 10 Washer Pump/Motor

1 - MOTOR
2 - CONNECTOR RECEPTACLE
3 - OUTLET NIPPLE
4 - INLET NIPPLE
5 - PUMP

rotor-type washer pump. A seal flange with a barbed inlet nipple on the pump housing passes through a rubber grommet seal installed in a dedicated mounting hole in the sump area near the bottom of the washer reservoir. On vehicles with the optional rear washer system, the front washer pump/motor unit is always mounted in the forward-most, lowest pump mounting hole of the reservoir. An integral filter screen is located within the pump inlet nipple. When the pump is installed in the reservoir a barbed outlet nipple on the pump housing connects the unit to the washer system through the front washer supply hose.

The washer pump/motor unit is retained on the reservoir by the interference fit between the barbed pump inlet nipple and the grommet seal, which is a light press fit. An integral connector receptacle between the motor housing and the pump housing connects the unit to the vehicle electrical system. A small capacitor internal to the connector receptacle insulator is connected in parallel with the motor leads to control ElectroMagnetic Interference (EMI) created by washer motor operation. The front washer pump/motor unit cannot be repaired. If faulty or damaged, the entire washer pump/motor unit must be replaced.

OPERATION

The front washer pump/motor unit features a small Direct Current (DC) electric motor. The motor is connected to the vehicle electrical system through a single take out and two-cavity connector of the headlamp and dash wire harness. The motor is grounded at all times through another take out of the headlamp and dash wire harness with a single eyelet terminal connector that is secured under a ground screw to the radiator closure panel behind the left headlamp in the engine compartment. The motor receives battery current through the closed contacts of the momentary front washer switch circuitry within the right multi-function switch only when the switch control stalk is pulled towards the steering wheel.

Washer fluid is gravity-fed from the washer reservoir to the inlet side of the washer pump. When the pump motor is energized, the motor spins the rotor within the rotor-type washer pump. The spinning pump rotor pressurizes the washer fluid and forces it through the pump outlet nipple, the front washer plumbing, and the front washer nozzle onto the windshield glass. The front washer pump/motor unit can be diagnosed using conventional diagnostic tools and methods.

REMOVAL

(1) Disconnect and isolate the battery negative cable.

(2) Disconnect the washer supply hose from the barbed outlet nipple of the front washer pump/motor and allow the washer fluid from the washer reservoir to drain into a clean container for reuse (Fig. 11).

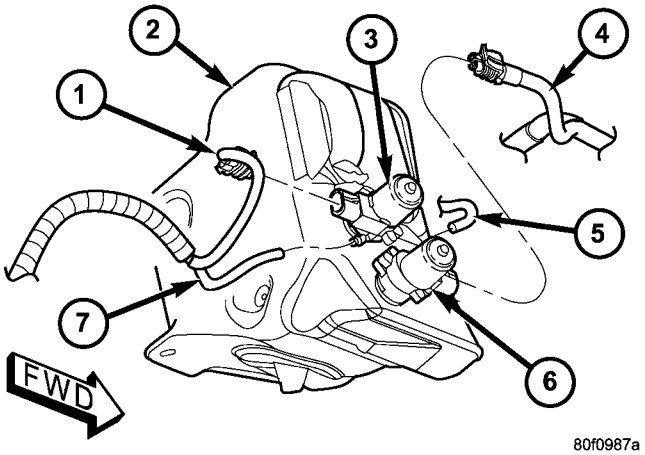

Fig. 11 Washer Pump/Motor Remove/Install

1 - REAR BODY WIRE HARNESS
2 - WASHER RESERVOIR
3 - REAR WASHER PUMP/MOTOR
4 - HEADLAMP & DASH WIRE HARNESS
5 - FRONT WASHER SUPPLY HOSE
6 - FRONT WASHER PUMP/MOTOR
7 - REAR WASHER SUPPLY HOSE

(3) Disconnect the headlamp and dash wire harness connector from the connector receptacle for the front washer pump/motor.